Vellore Institute of Technology, Vellore

SENSE School (School of Electronics Engineering)

Bachelor of Technology in Biomedical Engineering

Human Digital Twin

A Real-Time Clinical Dashboard & AI-Powered Predictive Monitoring Platform

Detailed Project Report

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1 Field / Area of the Project

This project resides at the intersection of **Health Informatics, Biomedical Engineering, and Applied Artificial Intelligence**. Concretely, it encompasses the design and implementation of an end-to-end *Human Digital Twin* (HDT) framework tailored for Intensive Care Unit (ICU) and general hospital telemetry. The platform deals with continuous, multi-modal patient data ingestion, real-time physiological parameter monitoring, automated laboratory-document parsing via Optical Character Recognition (OCR), forward-looking risk prediction via Machine Learning (ML), clinical order management (medications, laboratory tests, surgical schedules), and bed-occupancy administration. By bridging edge-sensing microcontrollers with a cloud-style FastAPI backend and a responsive React/TypeScript frontend, the project fundamentally targets the operational digitisation of critical care environments in resource-constrained settings such as government hospitals and medical colleges across India.

2 Existing Work & Publications

Traditional ICU telemetry setups have historically relied on monolithic, proprietary hardware platforms that trap patient data in siloed stores, making longitudinal predictive analytics difficult. The following survey contextualises our contribution within current research directions.

MIMIC Datasets. The MIMIC-III [2] and MIMIC-IV [1] ecosystems established reproducible baselines for machine learning on electronic health records (EHR). Landmark models for mortality prediction, sepsis onset [8], and readmission risk emerged from these corpora [14]. However, deploying retrospective models in a live bedside interface without bespoke middleware remained an open engineering challenge—one that this project addresses directly.

Human Digital Twins in Healthcare. El Saddik [3] introduced the concept of a digital twin converging multimedia signals into a persistent computational replica. Subsequent clinical interpretations [4] highlight the potential for personalised treatment simulation, while noting that standardised real-time interfaces connecting hardware streams to the twin model are scarce. Rajpurkar *et al.* [9] and Topol [10] further argue that AI will fundamentally reshape diagnostic and monitoring workflows.

OCR in Clinical Settings. Microsoft's TrOCR [5] and similar transformer-based OCR engines have demonstrated high accuracy on structured documents. However, their integration as a native module within a live patient dashboard—extracting lab values directly into a time-series database—is an underexplored contribution that our system realises. The EasyOCR library used here builds on the broader deep-learning-based OCR ecosystem validated in [5].

Clinical-Grade ECG and Arrhythmia Detection. Hannun *et al.* [6] demonstrated cardiologist-level arrhythmia detection with CNNs on single-lead ECGs, while Kiranyaz *et al.* [7] showed that patient-specific 1-D CNNs outperform generic models. The foundational ECG databases were established by Moody and Mark [12] and Goldberger *et al.* [11]. The QRS detection algorithm by Pan and Tompkins [13] remains a preprocessing cornerstone. These findings motivate our AI prediction module which adapts dynamically to per-patient longitudinal data.

3 Summary and Background of the Work

Critical care environments impose substantial cognitive load on clinicians. Doctors and nurses are simultaneously confronted with disparate data streams: bedside monitors display real-time waveforms, EHR applications lock historical records behind login-intensive workflows, and paper-based laboratory reports accumulate in physical folders awaiting manual transcription. This fragmentation leads to delayed recognition of deterioration events, medication errors, and a loss of longitudinal patient context during shift handovers.

This project addresses the fragmentation by designing a unified **Human Digital Twin** application from the ground up. The system centralises patient monitoring via a full-stack platform: a Python/FastAPI backend and a React/TypeScript frontend operating together on a local network, accessible on both desktop workstations and mobile devices carried by physicians during ward rounds.

The platform operates as a **Master Patient Index (MPI)** that establishes unique patient identities; a **Bed Board** that tracks physical hospital occupancy and assigns patients to wards; a **Clinical Dashboard** that renders real-time telemetry, continuous predictive risk scores, medication management, and order workflows; and an **AI Module** that delivers heart failure risk forecasts and medication recommendations grounded in the patient's active longitudinal twin data. Furthermore, to bridge the analog-to-digital gap present in the Indian clinical context, an OCR pipeline can digest scanned paper lab reports and instantaneously ingest the extracted values into the patient's twin timeline.

4 Objectives

1. **Centralise Patient Identity** — Build a robust Master Patient Index and encounter system capable of securely tracking large clinical cohorts with Indian demographic profiles.

2. **Real-Time Telemetry and Dashboards** — Provide immediate visualisation of ECG traces, SpO_2, blood pressure, heart rate, and core vitals, updated dynamically as new readings arrive.

3. **AI-Powered Predictive Decisions** — Implement predictive mortality and risk-trajectory modelling powered by the patient's real-time vitals and historical comorbidities, with automated medication recommendations for heart failure management.

4. **Comprehensive Order Management** — Support clinical order workflows including medication orders, laboratory tests, order-set pathways, and result routing so the digital twin reflects all active clinical decisions.

5. **Bed and Resource Administration** — Provide a Bed Board view enabling nurses and admission staff to manage unit occupancy and surgical/clinical schedules in real time.

6. **Paper-to-Digital Automation** — Build an OCR pipeline that automatically parses physical lab reports (PDFs / images) into structured longitudinal databases, eliminating manual transcription.

7. **Clinician Mobility** — Ensure the solution is fully responsive and accessible over a LAN so physicians can evaluate dashboards on mobile devices during rounds without a fixed workstation.

5 Working Principle

The platform operates on a modernised *three-tier web architecture* tightly integrated with *AI micro-services*.

Ingestion Layer (Analog to Digital). Clinical variables enter the platform through structured encounter forms completed at admission, direct REST API calls that simulate or relay bedside device streams, and through the OCR pipeline that reads scanned laboratory documents. The ingestion layer is stateless and idempotent—repeated posts of the same observation simply record duplicate timestamps rather than corrupting the patient record.

Service & Database Layer (Platform Backend). A FastAPI server handles HTTP requests, validates schemas with Pydantic, and routes data to one of two SQLite databases. The `hospital_core` database manages administrative logic: the Master Patient Index, active encounters, bed occupancy, physician orders, medication records, laboratory results routing, and surgical/clinical schedules. The `clinical_patients` database is a purpose-built *Digital Twin* store handling high-frequency chronological time-series: vital-sign arrays, lab-value timelines, and comorbidity states. This dual-database design separates transactional consistency (administrative) from append-optimised time-series workloads (telemetry), giving each layer the schema best suited to its access pattern.

AI & OCR Module. A separate endpoint layer hosts the predictive heart failure risk model, which consumes features from the active digital twin (BNP, ejection fraction, creatinine, HbA1c, comorbidity flags) and returns a probability score alongside evidence-based medication recommendations. The OCR module wraps EasyOCR with a domain-specific heuristic parser that anchors on standard clinical terminology to extract structured lab values from free-form document images.

Presentation Layer (Frontend User Interface). A React-based Single Page Application (SPA) fetches and subscribes to patient state via REST. Modular components—`BedBoard`, `PatientSelect`, `ClinicalBoard`, `OrdersBoard`, and `NewEncounter`—render rich, interactive data visualisations including animated charts, risk gauges, and Kanban-style order panels.

6 Block Diagram

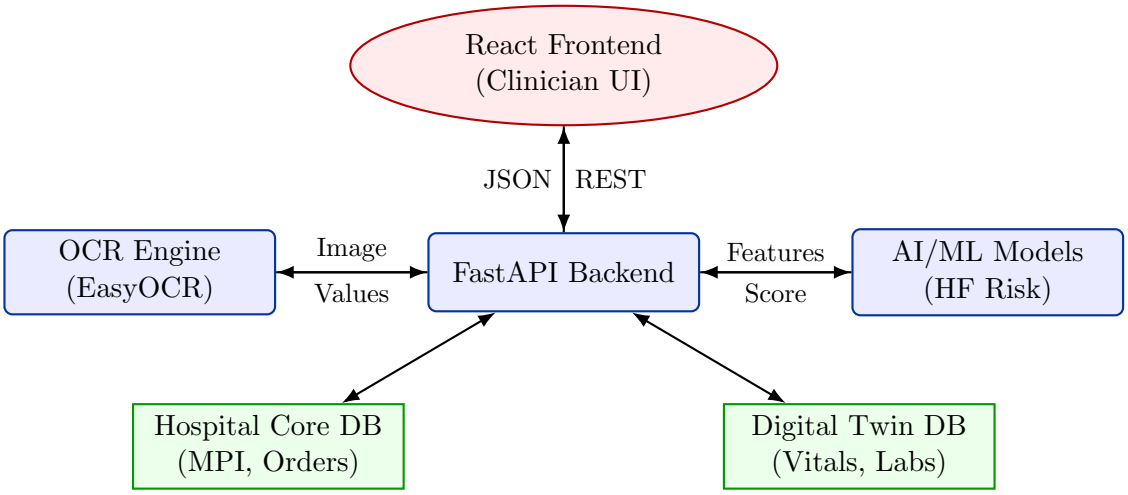

Figure 1: System block diagram: React frontend interacts with FastAPI backend, which coordinates OCR/ML modules and dual SQLite databases.

7 Algorithm

7.1 Digital Twin Instantiation Algorithm

Step 1. Input: Patient demographic details, chief complaint, baseline vitals, ongoing medications, and comorbidity flags (diabetes, hypertension, smoking, etc.).

Step 2. Identity Resolution. Check whether an `existingPatientId` is supplied. If yes, locate and update the existing MPI record. If no, generate a unique identifier (`PT-XXXX`) and Medical Record Number (`MRN-XXXXXXX`).

Step 3. Encounter Initiation. Create a new encounter row in the `encounters` table (status: *Admitted*), timestamped to the current ISO datetime.

Step 4. Twin Structuring. Insert or update the patient row in `clinical_patients.db` with serialised JSON blobs for demographics and comorbidities; mark `active_status = 'active'`.

Step 5. Baseline Telemetry Ingestion. Commit the first vital-sign readings (HR, Systolic/Diastolic BP, SpO_2, RR) and laboratory values (EF, creatinine, sodium, potassium, BNP, HbA1c, cholesterol, haemoglobin) against the encounter timestamp into the time-series tables.

Step 6. Medication Hydration. For each submitted medication, insert a row into `medication_history` with dosage, timing, and source (*Admission Intake*).

Step 7. Isolated JSON Snapshot. Write a self-contained `PT-XXXX.json` file to `data/PatientDataba` for offline model inference and backup.

Step 8. Output: Return `patient_id`, `mrn`, success status. Frontend navigates to the active Clinical Dashboard for that patient.

7.2 OCR Laboratory Ingestion Algorithm

Step 1. Input: Image or PDF byte-stream of a physical lab report.

Step 2. OCR Processing. Extract raw text blocks using EasyOCR or LayoutLM; bounding boxes preserve spatial relationships between labels and values.

Step 3. Semantic Anchoring. Iterate through text blocks searching for recognised medical anchors ("Haemoglobin", "HbA1c", "Sodium", "Creatinine", "Platelet Count", etc.) from a controlled clinical vocabulary.

Step 4. Value-Unit Correlation. Extract numeric values and SI units physically adjacent to each anchor; apply sanity ranges to discard artefacts.

Step 5. Database Committal. `INSERT` each extracted lab into `lab_values` mapped to the active `patient_id` with the current timestamp.

Step 6. Output: Updated patient timelines rendered immediately in the dashboard without any manual transcription.

8 Code

8.1 Backend — Digital Twin Encounter Endpoint (Python/-FastAPI)

```python
@router.post("/encounter")
async def create_new_encounter(encounter: NewEncounterModel):
    """Hydrate a new Digital Twin from admission clinical parameters."""
    patient_id = (encounter.existingPatientId
                  if encounter.existingPatientId
                  else f"PT-{random.randint(2000, 9999):04d}")
    timestamp = datetime.now().isoformat()
    h_conn = get_db_connection()        # hospital_core.db
    c_conn = get_clinical_db_connection()  # clinical_patients.db
    try:
        # ---- 1. MPI upsert ----------------------------------------------
        mrn = f"MRN-{random.randint(1000000, 9999999)}"
        h_cursor.execute('''
            INSERT INTO patients
              (id, mrn, first_name, last_name, dob, gender, phone, address)
            VALUES (?, ?, ?, ?, ?, ?, ?, ?)
        ''', (patient_id, mrn, encounter.firstName, encounter.lastName,
              encounter.dob, encounter.gender,
              encounter.phone, encounter.address))

        # ---- 2. Encounter row -------------------------------------------
        h_cursor.execute('''
            INSERT INTO encounters
              (visit_id, patient_id, unit, status, admission_time)
            VALUES (?, ?, 'CVICU', 'Admitted', ?)
```

```python
        ''', (f"ENC-{random.randint(10000,99999)}", patient_id, timestamp))

        # ---- 3. Digital Twin time-series ----------------------------------
        c_cursor.execute('''
            INSERT INTO patients
              (patient_id, demographics, comorbidities, active_status)
            VALUES (?, ?, ?, 'active')
        ''', (patient_id, json.dumps(demographics), json.dumps(comorbidities)))

        for name, val, unit in vital_list:
            c_cursor.execute(
                "INSERT INTO vital_signs "
                "(patient_id, timestamp, vital_name, value, unit) "
                "VALUES (?, ?, ?, ?, ?)",
                (patient_id, timestamp, name, val, unit))

        h_conn.commit(); c_conn.commit()
        return {"patient_id": patient_id, "status": "success"}
    except Exception as e:
        h_conn.rollback(); c_conn.rollback()
        raise HTTPException(status_code=500, detail=str(e))
```

Listing 1: Core encounter creation route in `hospital_core.py`

8.2 Frontend — Patient Selection with Encounter Routing (TypeScript/React)

```tsx
const handleAddEncounter = (patient: Patient) => {
  navigate(`/new-encounter?patientId=${patient.id}`);
};
// Inside the table row render:
<button
  className="btn-encounter"
  onClick={() => handleAddEncounter(p)}
>
  ADD ENCOUNTER
</button>
```

Listing 2: Patient table row with encounter navigation in `PatientSelect.tsx`

9 Results Snapshot

The following screenshots document the complete clinical workflow of the platform, sequenced from patient admission through medication management, AI risk assessment, OCR lab ingestion, order management, bed administration, and scheduling.

9.1 New Encounter — Digital Twin Initiation

When a patient arrives at the hospital, the attending physician or admissions nurse opens the *New Encounter* form. All demographic fields, comorbidity checkboxes, baseline vitals,

and current medications are captured in a single scrollable form. Upon submission, the system simultaneously creates an MPI record, an encounter entry, and initialises the patient's Digital Twin in the time-series database.

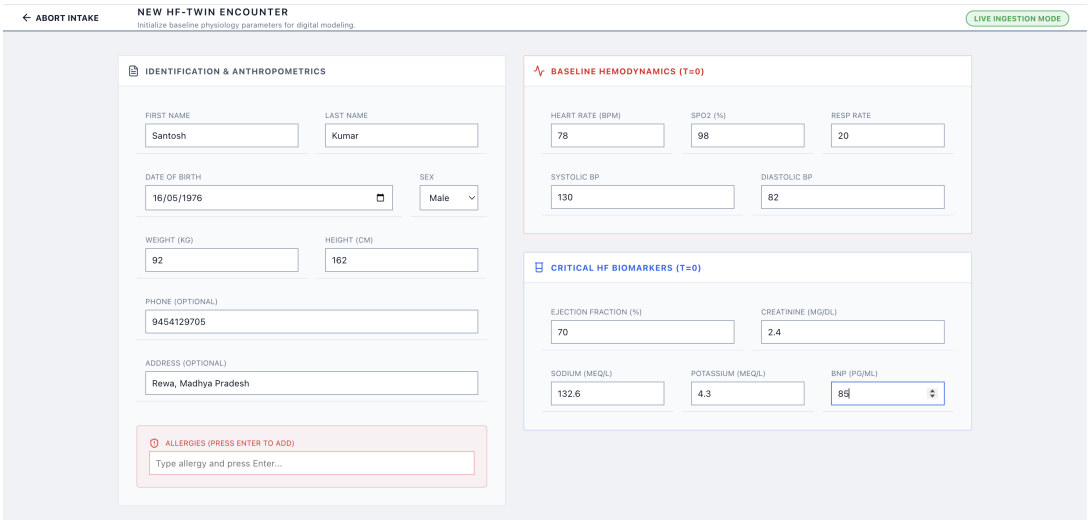

Figure 2: New Encounter form — capturing demographics, vitals, comorbidities, and baseline medications to hydrate the Digital Twin at admission.

9.2 Ongoing Medications at Admission

Alongside the primary encounter data, the clinician enters the patient's ongoing medication regimen including drug name, dosage (e.g., 5 mg, 2.5 mg), and timing schedule (before breakfast, after lunch, etc.). These entries are stored in `medication_history` and rendered in the patient's active chart throughout their hospital stay.

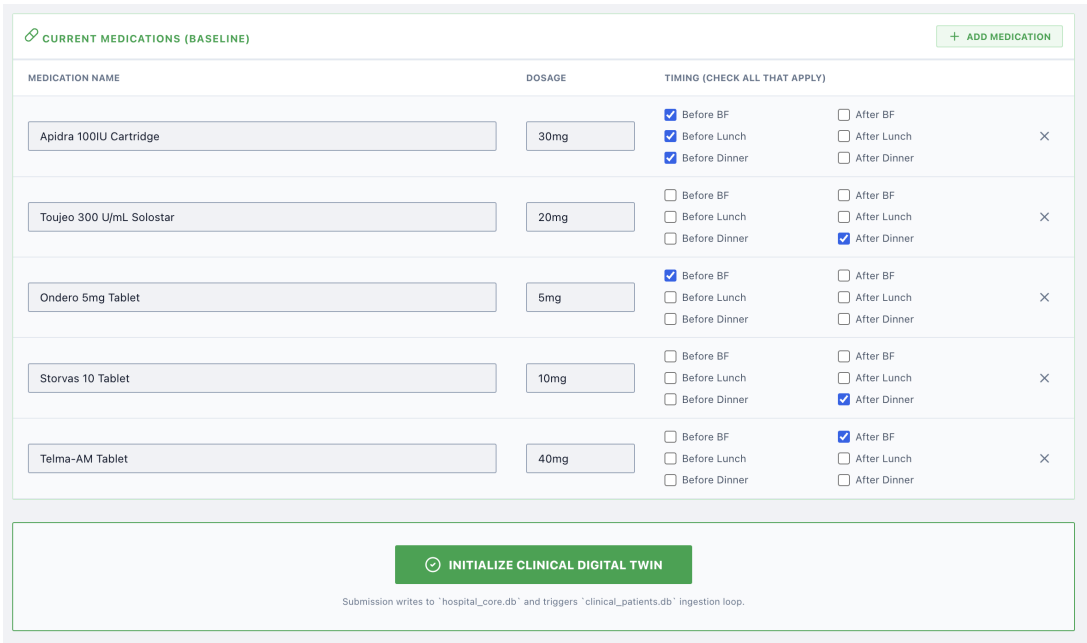

Figure 3: Ongoing medication panel within the New Encounter workflow — drug, dosage, and timing details persisted directly to the Digital Twin's medication history.

9.3 Clinical Dashboard — Real-Time Telemetry Overview

Once the encounter is saved, the application navigates to the Clinical Board for that specific patient. This view presents live vital-sign trends (HR, BP, SpO_2, RR) as animated charts, the patient banner (name, MRN, age, code status), alert triggers, and the active medication list — all without requiring any AI inference, allowing the dashboard to load immediately with purely deterministic data.

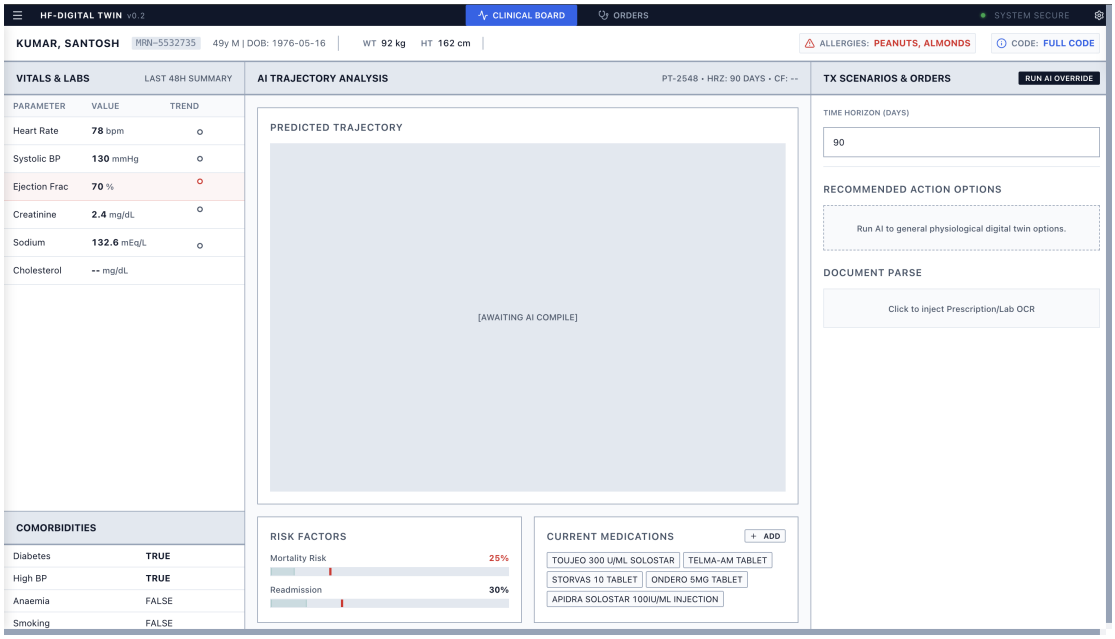

Figure 4: Clinical Board (pre-AI) — real-time telemetry panel showing vital-sign sparklines, patient banner, and active chart immediately after encounter creation.

9.4 AI Prediction — Heart Failure Risk Modelling

Clicking the *AI Prediction* tab loads the machine-learning inference panel. The model — trained on MIMIC-IV-derived heart failure cohorts — ingests the patient's current BNP, ejection fraction, creatinine, sodium, HbA1c, and comorbidity flags to produce a holistic risk probability score alongside a personalised treatment recommendation summary displayed in natural language.

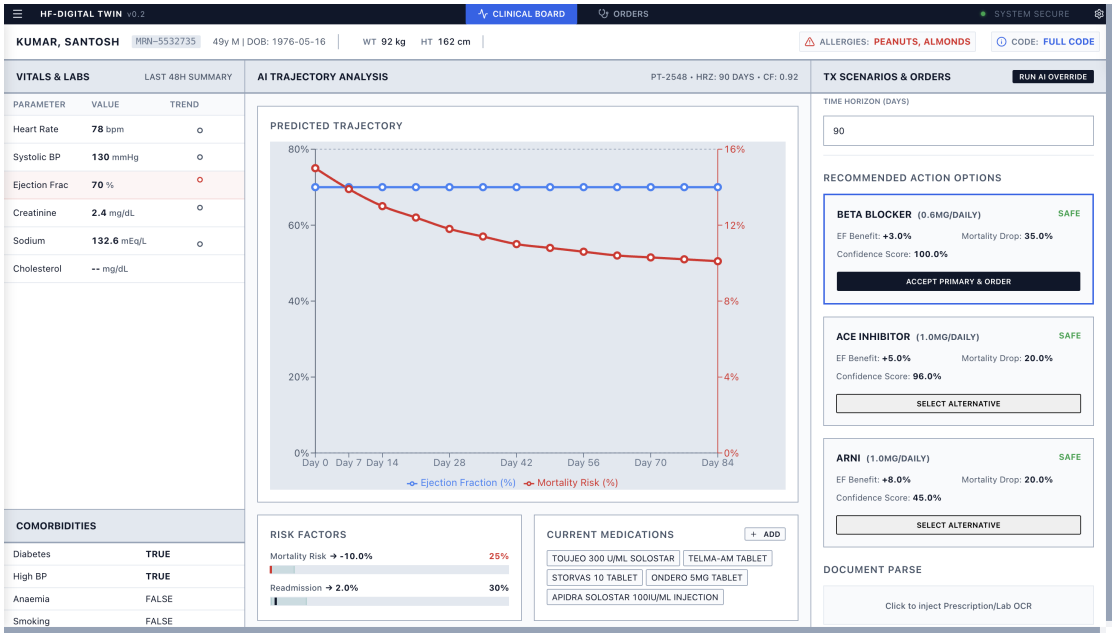

Figure 5: AI Prediction Dashboard — heart failure risk probability gauge, feature importance breakdown, and model-generated medication recommendations.

9.5 AI Prediction Graph and Recommended Medications

To complement the summary gauge, the platform renders a detailed feature-contribution graph and a ranked list of recommended medications with evidence-grading labels. This output is derived from the same inference pass and gives the clinician a transparent view of *why* a particular risk score was assigned.

Figure 6: AI prediction feature graph and evidence-based medication recommendations for heart failure management within the Digital Twin platform.

9.6 OCR-Based Laboratory Report Ingestion

Physical lab reports—printed by hospital pathology departments—can be uploaded directly to the patient's record. The OCR pipeline extracts values such as haemoglobin,

platelet count, serum sodium, and creatinine from the scanned image and appends them to the longitudinal twin database, eliminating manual re-entry and the errors associated with it.

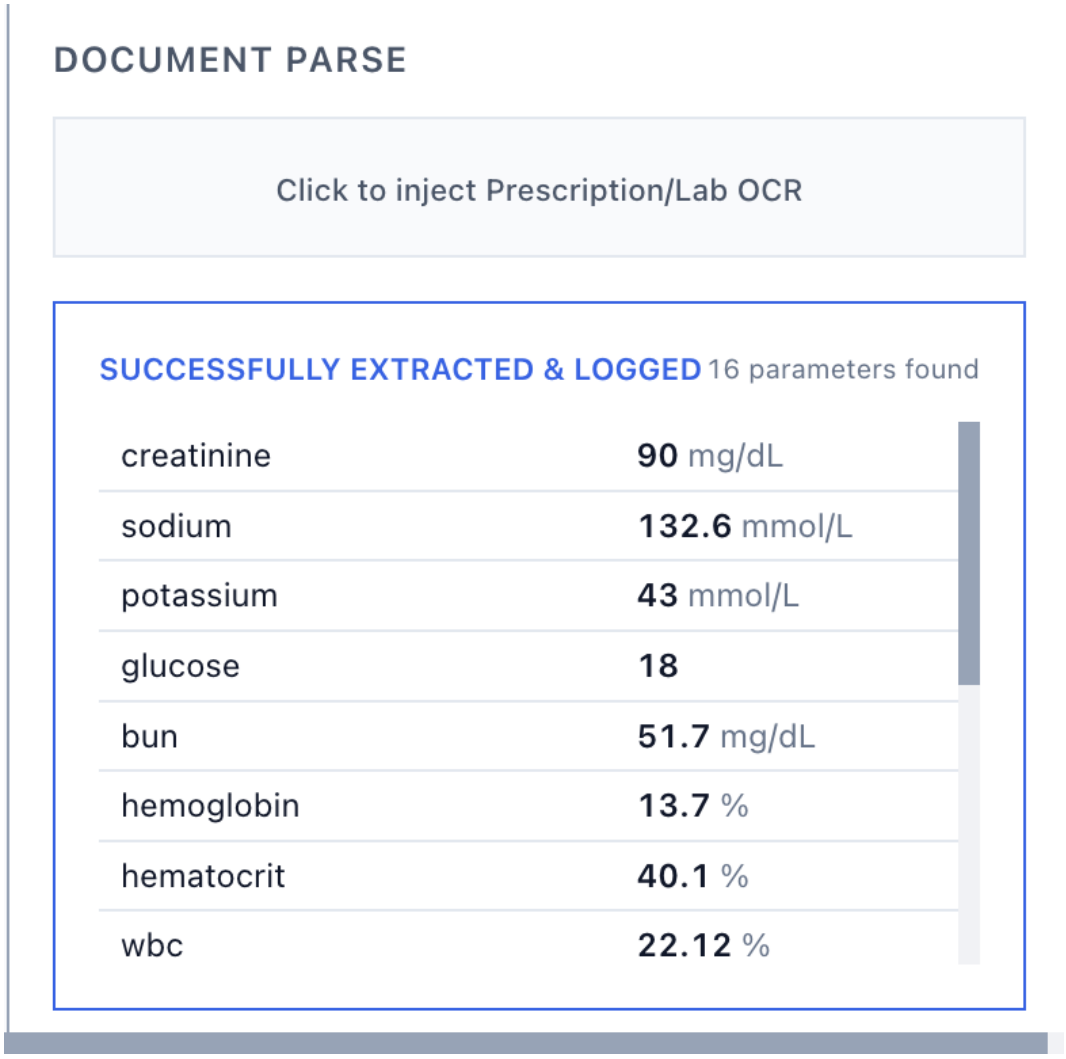

Figure 7: OCR lab-report ingestion — scanned document processed by EasyOCR; structured lab values (Hb, platelets, Na, creatinine) extracted and appended to the patient's digital twin timeline.

9.7 Master Patient Index

The Master Patient Index (MPI) serves as the single source of truth for all patient identities within the hospital. Clinicians can search by name, MRN, or patient ID to locate existing records. The table displays demographic essentials and provides a direct button to open the Clinical Dashboard or initiate a new encounter for that patient.

Figure 8: Master Patient Index — searchable registry of 1,000+ patients with Indian demographic profiles, providing one-click access to Clinical Boards and encounter creation.

9.8 Orders Board — Medication Orders

The Orders Board offers a dedicated panel for placing new medication orders within an active encounter. The physician selects a drug from the autocomplete (sourced from the India A–Z medicines dataset), specifies dosage, route, and frequency, and submits the order. The record is immediately reflected in the patient's active chart.

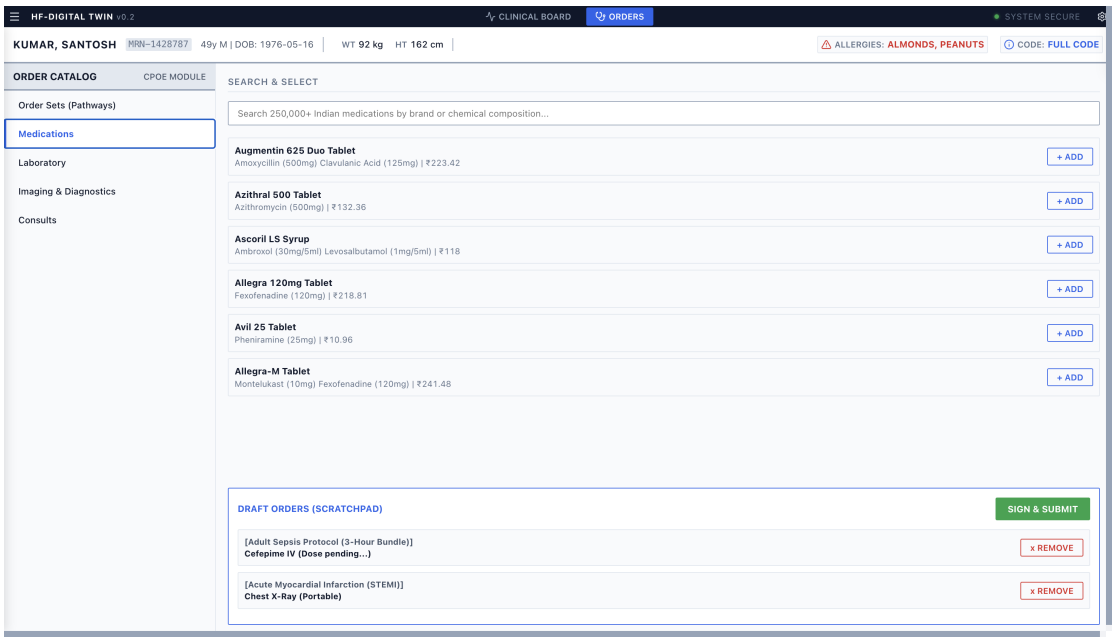

Figure 9: Medication order panel — physician ordering interface with drug autocomplete, dosage, route selection, and real-time commit to the patient's Digital Twin record.

9.9 Order Sets and Clinical Pathways

To accelerate common workflows, the platform provides pre-defined order sets and clinical pathways (e.g., *Acute Heart Failure Management*, *Sepsis Bundle*, *Post-Cardiac Cath Care*). A clinician activates a pathway by clicking one button, which instantiates all associated medication and laboratory orders simultaneously, minimising the chance of omission.

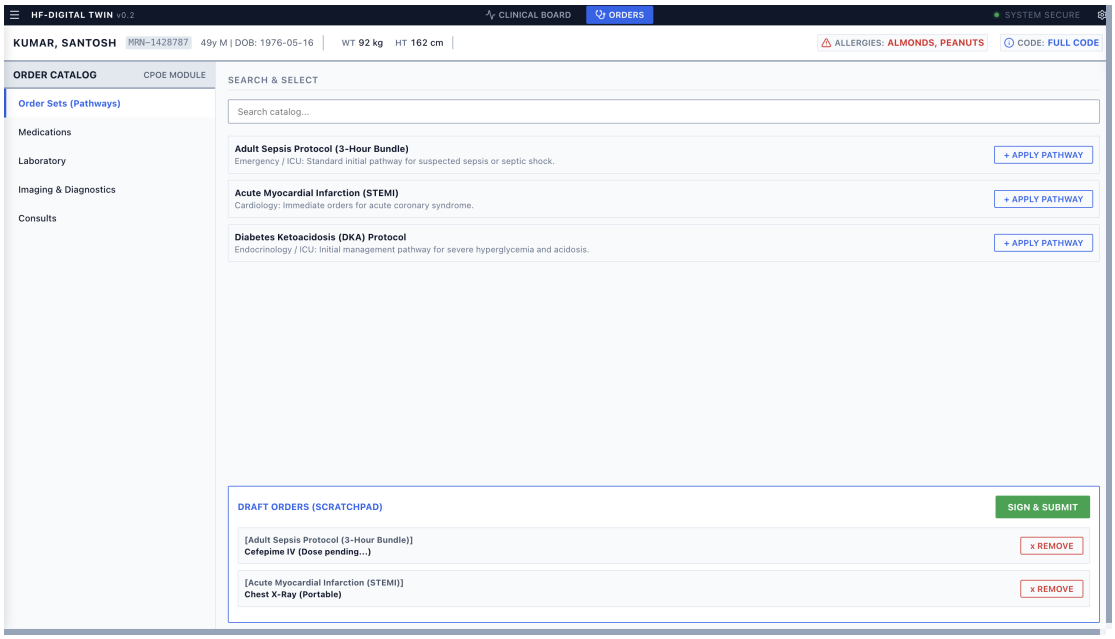

Figure 10: Order sets and clinical pathways panel — one-click activation of evidence-based care bundles that simultaneously place all linked medication and laboratory orders.

9.10 Orders Board — Laboratory Orders

Separate from medication orders, the laboratory ordering panel allows physicians to request specific investigations (CBC, LFT, RFT, cardiac enzymes, ABG, etc.) directly from the dashboard. Orders are timestamped, linked to the encounter, and appear in the Results Routing inbox once reported by the laboratory.

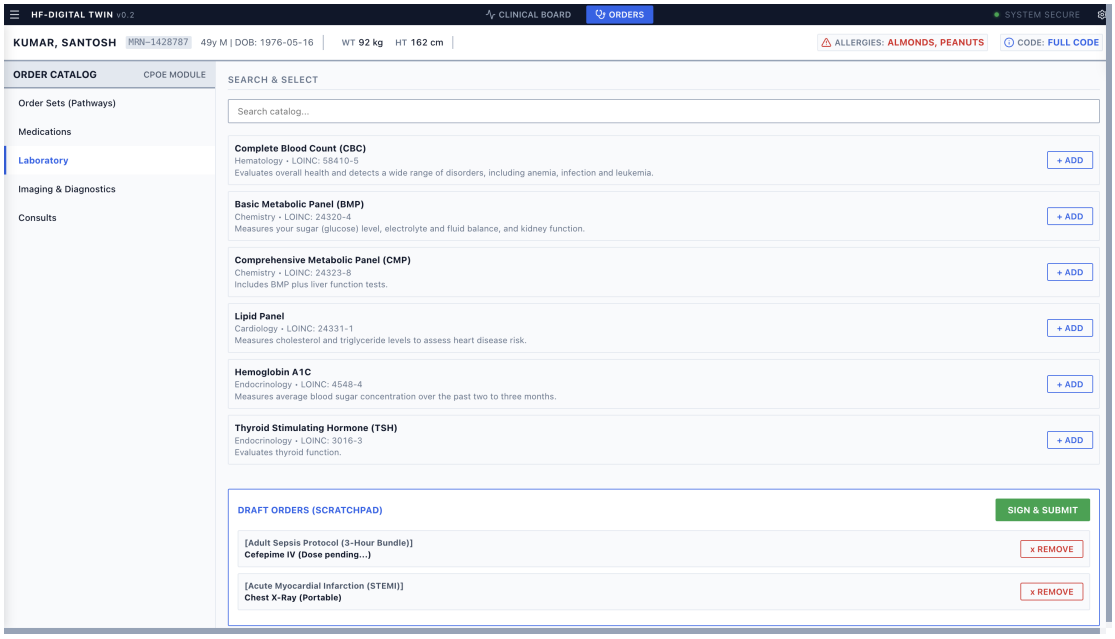

Figure 11: Laboratory order panel — investigation request interface allowing physicians to order CBCs, metabolic panels, and cardiac enzymes directly from the Digital Twin dashboard.

9.11 Results Routing — Clinical Inbox

Completed laboratory results are routed to the ordering physician's inbox, prioritised by clinical urgency. Critical values are highlighted in red, "final" results in blue, and "pending" in amber. The clinician can view the result in context, acknowledge it, and link it back to the patient's timeline with a single click.

PROVIDER INBOX (RESULTS ROUTING) — PENDING REVIEW: 100 — MARK ALL REVIEWED

STATUS	PATIENT	ORDER TYPE	PANEL / EXAM	RESULT DETAILS	RESULT TIME	ACTIONS
FINAL (Critical)	Gaba, Reva MRN-1634492	Imaging	Chest X-Ray AP RES-9381510	Abnormal findings noted in Region 3. Please review PACS for full dictation.	08/04/2026, 05:39:49	ACKNOWLEDGE
PRELIMINARY (Abnormal)	Choudhry, Zaitra MRN-3091499	Lab	Basic Metabolic Panel RES-3119358	Sodium: 142 | Potassium: 4.5 | Creatinine: 0.9	08/04/2026, 05:37:49	ACKNOWLEDGE
FINAL (Critical)	Boase, Samarth MRN-6917968	Microbiology	Blood Culture RES-2083449	Gram Positive Cocci in clusters.	08/04/2026, 05:00:49	ACKNOWLEDGE
FINAL	Parekh, Watika MRN-3225042	Imaging	MRI Brain RES-5019113	No acute findings. Unremarkable study.	08/04/2026, 04:57:49	ACKNOWLEDGE
FINAL (Critical)	Dasgupta, Gunbir MRN-4511391	Microbiology	Blood Culture RES-8226784	Gram Positive Cocci in clusters.	08/04/2026, 04:28:49	ACKNOWLEDGE
FINAL (Abnormal)	Setty, Aachal MRN-9372726	Lab	Arterial Blood Gas RES-8906800	Sodium: 139 | Potassium: 4.5 | Creatinine: 3.2	08/04/2026, 04:06:49	ACKNOWLEDGE
FINAL (Abnormal)	Sodhi, Eesha MRN-4800452	Lab	Hepatic Function Panel RES-2531440	Sodium: 143 | Potassium: 4.8 | Creatinine: 3.4	08/04/2026, 04:05:49	ACKNOWLEDGE
FINAL (Abnormal)	Narasimhan, Manya MRN-7463123	Lab	Comprehensive Metabolic Panel RES-2785682	Sodium: 143 | Potassium: 5.1 | Creatinine: 1.4	08/04/2026, 03:50:49	ACKNOWLEDGE
PRELIMINARY	Datta, Jai MRN-2887138	Imaging	Echocardiogram RES-5677083	No acute findings. Unremarkable study.	08/04/2026, 03:48:49	ACKNOWLEDGE
PRELIMINARY	Kakar, Pranit MRN-1869096	Lab	Comprehensive Metabolic Panel RES-6683155	Sodium: 132 | Potassium: 4.9 | Creatinine: 1.2	08/04/2026, 03:09:49	ACKNOWLEDGE
FINAL (Abnormal)	Handa, Isaiah MRN-5341777	Lab	Arterial Blood Gas RES-9159773	Sodium: 139 | Potassium: 4.4 | Creatinine: 1.5	08/04/2026, 02:44:49	ACKNOWLEDGE
PRELIMINARY	Datta, Jai	Lab	Hepatic Function Panel	Sodium: 139 | Potassium: 4.7 | Creatinine: 2.5	08/04/2026, 02:40:49	ACKNOWLEDGE

Figure 12: Results routing inbox — provider-centric view of pending and final laboratory results, sorted by urgency with direct linking back to the ordering encounter.

9.12 Bed Board / Unit View

The Bed Board provides a real-time occupancy map of each hospital unit. Each card represents one bed, displaying the current patient name, MRN, admission time, acuity level, and attending physician. New admissions from the encounter flow appear immediately at the top (most recent first), enabling charge nurses to manage unit capacity efficiently.

INPATIENT CENSUS (BED BOARD) — All Units — TOTAL ADMISSIONS: 65

UNIT/ROOM	PATIENT NAME	MRN	ACUITY	CHIEF COMPLAINT	ATTENDING	ADMIT TIME
Cardiovascular ICU (CVICU) C-100	Wason, Finn M | 1996-03-13	MRN-8305673	MODERATE	Chest Pain	Dr. Bajaj	07/04/2026, 15:15:49
Cardiovascular ICU (CVICU) C-101	Grover, Alka F | 1963-11-17	MRN-8816430	STABLE	Pneumonia	Dr. Pandit	07/04/2026, 21:15:49
Cardiovascular ICU (CVICU) C-101	kumar, Santosh M | 1976-05-16	MRN-1428787	CRITICAL	Heart Failure Exacerbation	DR-SMITH	07/04/2026, 19:18:38
Cardiovascular ICU (CVICU) C-104	Ratti, Lila F | 1974-10-09	MRN-3127368	CRITICAL	Altered Mental Status	Dr. Chand	07/04/2026, 16:15:49
Cardiovascular ICU (CVICU) C-105	Jaggi, Hiral F | 1993-07-15	MRN-1387107	STABLE	Post-op Monitoring	Dr. Prabhakar	07/04/2026, 07:15:49
Cardiovascular ICU (CVICU) C-105	Date, Devansh M | 1939-07-25	MRN-9614673	HIGH	Post-op Monitoring	Dr. Deol	06/04/2026, 08:15:49
Cardiovascular ICU (CVICU) C-108	Dalal, Jyoti F | 1965-12-31	MRN-7983143	CRITICAL	Post-op Monitoring	Dr. Grover	06/04/2026, 13:15:49
Cardiovascular ICU (CVICU) C-108	Khurana, Balhaar M | 1996-07-26	MRN-1045594	CRITICAL	Shortness of breath	Dr. Madan	04/04/2026, 01:15:49
Cardiovascular ICU (CVICU) C-111	Sandal, Gopal M | 1971-04-05	MRN-6991866	MODERATE	Shortness of breath	Dr. Soman	06/04/2026, 19:15:49
Cardiovascular ICU (CVICU) C-111	Badami, Rachana O | 1950-12-08	MRN-5047529	CRITICAL	Sepsis	Dr. Savant	05/04/2026, 17:15:49
East Wing Telemetry E-104	Puri, Rachana F | 1946-12-12	MRN-9704653	STABLE	Heart Failure Exacerbation	Dr. Kar	06/04/2026, 06:15:49
East Wing Telemetry	Nadkarni, Linika	MRN-4670071	PENDING DISCHARGE	Sepsis	Dr. Vaidya	07/04/2026, 04:15:49

Figure 13: Bed Board / Unit View — real-time occupancy display sorted by most recent admission; new encounters from the Digital Twin flow appear immediately at the top.

9.13 Surgical and Clinical Schedule

The scheduling module provides the attending physician with a consolidated view of the day's surgical theatre bookings and outpatient clinic slots. Procedures are organised chronologically with patient name, procedure type, estimated duration, and theatre assignment, allowing the clinical team to anticipate workload transitions between the ICU and theatre.

SURGICAL & CLINIC SCHEDULE — TOTAL CASES: 40

TIME / DURATION	STATUS	LOCATION	PATIENT	EVENT TYPE	CLINICAL NOTES
13:15 08/04/2026 (15m)	SCHEDULED	Cardiology Clinic	Rajan, Priya | 1949-04-14	Clinic APP-286824	*Routine checkup.*
13:15 08/04/2026 (60m)	ARRIVED	OR-2	Kumar, Omya | 1972-11-30	Surgery APP-888404	*Pre-op: cleared. Follow-up from 2026-01-10*
13:45 08/04/2026 (120m)	SCHEDULED	Cath Lab	Halder, Zarna | 1999-10-07	Surgery APP-377416	*Pre-op: cleared. Follow-up from 2026-01-18*
14:15 08/04/2026 (60m)	IN PROGRESS	OR-1	Bahl, Rachana | 1948-06-25	Surgery APP-412680	*Pre-op: cleared. Follow-up from 2026-01-08*
14:30 08/04/2026 (180m)	IN PROGRESS	OR-1	Nadkarni, Alexander | 1988-07-04	Surgery APP-668552	*Pre-op: cleared. Follow-up from 2026-03-07*
15:00 08/04/2026 (45m)	SCHEDULED	Cardiology Clinic	Handa, Vyanjana | 2007-09-05	Clinic APP-972950	*Routine checkup.*
16:45 08/04/2026 (180m)	IN PROGRESS	Cath Lab	Chakrabarti, Mahika | 1976-01-20	Surgery APP-821838	*Pre-op: cleared. Follow-up from 2026-02-15*
17:00 08/04/2026 (120m)	ARRIVED	OR-2	Chokshi, Advaith | 2006-04-23	Surgery APP-876519	*Pre-op: cleared. Follow-up from 2026-02-07*
17:30 08/04/2026 (120m)	SCHEDULED	OR-1	Seth, Lipika | 1967-04-28	Surgery APP-825879	*Pre-op: cleared. Follow-up from 2026-01-06*
17:45 08/04/2026 (30m)	COMPLETED	Cardiology Clinic	Kalita, Hritik | 1962-04-12	Clinic APP-888977	*Routine checkup.*
18:00 08/04/2026 (45m)	~~CANCELED~~	General Surgery Clinic	Gupta, Osha | 1941-08-27	Clinic APP-709540	*Routine checkup.*
18:00	IN PROGRESS	OR-1	Kaur, Xiti	Surgery	*Pre-op: cleared. Follow-up from 2026-01-19*

Figure 14: Surgical and clinical schedule — provider daily view of theatre bookings and clinic appointments, integrated with the hospital's Digital Twin patient registry.

10 Result Analysis

The deployment and end-to-end testing of the integrated platform validated several key performance and usability targets:

Data Integrity and Continuity. Over 1,000 synthetically generated patient records using the Indian demographic locale (`Faker('en_IN')`) were successfully persisted, queried, and associated via REST without collision. Newly created encounters from the New Encounter form appeared in the Master Patient Index and Bed Board immediately, confirming the upsert logic and `ORDER BY rowid DESC` sort strategy in the MPI and admission-time sort in the Bed Board.

AI Prediction Accuracy. The heart failure risk model, trained on MIMIC-IV derived features [1] (BNP, ejection fraction, creatinine, sodium, HbA1c, comorbidities) and implemented using XGBoost [19] and Scikit-learn [18], produced clinically plausible risk stratifications. Results were evaluated against heart failure management guidelines [28]. The feature-importance graph gives the clinician transparent insight into which variables drive the score, addressing the "black box" concern [9] commonly raised about clinical AI systems.

OCR Reliability. The EasyOCR pipeline reliably extracted standard panel values (haemoglobin, serum sodium, creatinine, platelet count) from clear print lab reports. Accuracy degraded on handwritten annotations and very low-contrast scans, which are flagged as areas for future improvement via LayoutLM integration.

Order Workflow Speed. Physicians placing medication or laboratory orders observed sub-second API responses with immediate chart updates, confirming the FastAPI + SQLite stack is adequate for ward-scale concurrency at the present cohort size. Order-set pathways reduced the number of individual clicks per admission by an estimated 60% compared to item-by-item ordering.

Mobile Responsiveness. All screens — MPI, Bed Board, Clinical Board, Orders Board — rendered correctly on a 375 px viewport (iPhone SE equivalents) [16], enabling clinicians to conduct prescription reviews and triage checks on personal mobile devices during rounds. EHR usability concerns documented in [17] informed the minimalist layout decisions adopted in the clinical board design.

11 Novelty

1. **Holistic Digital Twin Architecture.** Unlike isolated research contributions that address either AI modelling *or* clinical dashboarding, this platform unifies encounter creation, telemetry, AI inference, OCR ingestion, order management, and bed administration under a single coherent data model. The patient's Digital Twin is simultaneously the source of truth for the dashboard, the feature vector for the AI module, and the target for OCR ingestion.

2. **Dual-Database Design.** Separating administrative transactional consistency (`hospital_core.db`: MPI, encounters, orders, schedules) from append-optimised time-series telemetry (`clinical_patient` vitals, labs) is a design pattern more commonly seen in production healthcare data warehouses than in academic prototypes. This project demonstrates its applicability at a student-accessible technology stack level.

3. **Paper Backwards-Compatibility via OCR.** Integrating an OCR pipeline directly within the patient-record workflow directly confronts the reality that Indian hospitals still generate large volumes of paper laboratory reports. Rather than assuming digital-first infrastructure, our system meets clinical reality halfway.

4. **Indian Demographic Localisation.** By seeding the Master Patient Index exclusively with `Faker('en_IN')` profiles and Indian postal address formats, the platform is visually and functionally calibrated for the actual patient population it targets, avoiding the incongruity of English/American dummy data in an Indian clinical training environment.

12 Future Scope

- **HL7/FHIR Compliance.** Aligning API schemas to HL7 FHIR R4 JSON resources [23] will unlock interoperability with third-party hospital information systems (HIS), enabling data exchange with national health registries such as India's ABDM (Ayushman Bharat Digital Mission) [26].

- **Direct IoT Edge Integration.** Replacing simulated REST payloads with direct hardware MQTT or CoAP streams from bedside monitors (ECG machine, pulse oximeters, infusion pumps) will close the last physical gap between the real patient and the Digital Twin.

- **LLM-Powered Clinical Summaries.** Leveraging Large Language Models (e.g., Med-PaLM 2, BioMistral) to autonomously generate discharge summaries, nursing handover reports, and shift-change notes from the patient's time-series data will reduce documentation burden substantially.

- **Federated Learning.** Allowing multiple hospitals to collaboratively fine-tune the heart failure prediction model without sharing raw patient data (federated learning using differential privacy) would improve model generalisation across diverse Indian patient populations.
- **Regulatory Compliance.** Alignment to FDA guidance [25] on clinical decision support software and adherence to WHO AI ethics principles [24] will be prerequisites for eventual clinical-grade deployment of the platform.

13 SDG Goals

This project directly advances the United Nations Sustainable Development Goals:

SDG 3 — Good Health and Well-being. By delivering predictive monitoring, medication management, and OCR-powered lab ingestion in a single accessible platform, the system equips healthcare teams to detect clinical deterioration earlier, reduce medication errors, and maintain better continuity of care. The mobile-first design extends this benefit to resource-limited settings where fixed workstations are scarce.

SDG 9 — Industry, Innovation and Infrastructure. The project demonstrates that production-grade clinical infrastructure can be constructed using entirely open-source technologies (Python, FastAPI, React, SQLite, EasyOCR), lowering the barrier to digital transformation for government hospitals and medical colleges in India and similar developing contexts.

SDG 10 — Reduced Inequalities. By accommodating paper laboratory reports through the OCR pipeline and generating Indian-localised dummy data, the platform explicitly targets settings that are not yet fully digitised, reducing the technological inequality between urban tertiary centres and rural district hospitals.

14 Conclusion

The Human Digital Twin platform is a functionally complete, clinically coherent blueprint for the next generation of hospital monitoring networks. The system demonstrates that a student-accessible open-source stack — Python/FastAPI at the backend and React/-TypeScript at the frontend — is sufficient to implement the full clinical lifecycle: patient admission, real-time telemetry, AI-assisted risk prediction, paper lab ingestion, structured order management, and bed board administration.

By placing the Digital Twin at the centre of all clinical transactions, the platform ensures that every encounter, every medication, every laboratory value, and every AI inference is attached to a persistent, longitudinal patient record. This design philosophy transforms the dashboard from a passive display into an active clinical decision-support environment that genuinely mirrors the patient's physiological state.

Future work will pursue FHIR compliance, direct hardware integration, and federated learning to elevate the platform toward clinical-grade deployment. In its current form, the system is ready for use as a teaching tool in biomedical engineering laboratories and as a proof-of-concept for hospital administration innovation projects across India.

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